

Math 526, F09
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	15	
5	10	
6	15	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3})$.

(a) Find the length of $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{3^2 + (\sqrt{3})^2 + (2\sqrt{6})^2} = \sqrt{36} = 6$$

$$\|\mathbf{w}\| = \sqrt{\mathbf{w} \cdot \mathbf{w}} = \sqrt{0^2 + (-\frac{\sqrt{3}}{3})^2 + (\frac{\sqrt{6}}{3})^2} = \sqrt{1} = 1$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{3 \cdot 0 + \sqrt{3} \cdot (-\frac{\sqrt{3}}{3}) + 2\sqrt{6} \cdot (\frac{\sqrt{6}}{3})}{6 \cdot 1}$$

$$= \frac{-1 + 4}{6} = \frac{1}{2}$$

$$\Rightarrow \theta = 60^\circ \text{ or } (\frac{\pi}{3})$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, \sqrt{3}, 2\sqrt{6})}{6} = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$$

(d) Find a number α so that $\mathbf{w} + \alpha\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$(\mathbf{w} + \alpha\mathbf{v}) \cdot \mathbf{v} = 0$$

$$\mathbf{w} \cdot \mathbf{v} + \alpha \|\mathbf{v}\|^2 = 0$$

$$\alpha = -\frac{\mathbf{w} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} = -\frac{3}{6^2} = -\frac{1}{12}$$

Problem 2. (20 points)

- (a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ x - y \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

- (b) Find an 3×3 elimination matrix $\mathbf{E}$ that adds 2 times row 1 to row 3.

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

- (c) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 2 and 3, then rows 1 and 2.

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- (d) Find two 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{0}$ but $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 3. (20 points)

(a) Write down the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it is invertible.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} &= \frac{1}{1 \cdot 4 - 2 \cdot 3} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

(b) Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find A^{-1} using the Gaussian-Jordan Elimination.

$$[A \ I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ \textcircled{1} & 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\ell_{21}=1} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ \textcircled{2} & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\ell_{31}=2} \left[\begin{array}{ccc|ccc} 1 & 1 & \textcircled{2} & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array} \right] \xrightarrow{\ell_{13}=2} \left[\begin{array}{ccc|ccc} 1 & \textcircled{1} & 0 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\ell_{12}=1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \end{array} \right] = [I \ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Problem 4. (15 points) Let $A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$ and $b = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix}$.

(a) Find the LU factorization of A.

$$A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=2} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{31}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} = U$$

Thus $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$

$\begin{matrix} \parallel & \parallel \\ L & U \end{matrix}$

(b) Using the above factorization to solve $Ax = b$ by solution of two triangular systems $Ly = b$ and $Ux = y$.

$$Ly = b \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix} \Rightarrow \begin{aligned} y_1 &= -6 \\ y_2 &= 4 + 2y_1 = -8 \\ y_3 &= -2 - 3y_1 - 3y_2 = -2 + 18 + 24 = 40 \end{aligned}$$

$$Ux = y \Rightarrow \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -6 \\ -8 \\ 40 \end{bmatrix}$$

$$\Rightarrow x_3 = \frac{40}{20} = 2$$

$$-2x_2 = -8 + 5x_3 = 2 \Rightarrow x_2 = -1$$

$$-x_1 = -6 - x_2 + 3x_3 = 1 \Rightarrow x_1 = -1$$

Thus $x = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$

Problem 5. (10 points) Find the column spaces of the following matrices.

(a) $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$.

$$Ax = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ 2x_2 \\ 0 \end{bmatrix}$$

$$\Rightarrow C(A) = \left\{ \begin{bmatrix} x_1 \\ 2x_2 \\ 0 \end{bmatrix} \mid x_1, x_2 \text{ are real numbers} \right\}.$$

$\rightarrow$ the x - y plane in $\mathbb{R}^3$

(b) $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$.

$$Bx = x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + (2x_2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= (x_1 + 2x_2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow C(B) = \left\{ c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \mid c \text{ is a real number} \right\}.$$

$\rightarrow$ the line spanned by the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Problem 6. (15 points) True or False (no explanation needed).

(a) If a matrix $\mathbf{A}$ is invertible, then $\mathbf{A}^3$ must be invertible.

True

(b) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, there is an 3×3 matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.

False

(c) The set of all vectors (b_1, b_2, b_3) with $b_1 \leq b_2 \leq b_3$ is a subspace of $\mathbb{R}^3$.

False

(d) If a subspace of the vector space $\mathbf{M}$ contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix $\mathbf{I}$.

True

(e) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric, then their product $\mathbf{AB}$ is symmetric.

False